

Supplementary Material: What Spatial Fine-Tuning Actually Changes

Anonymous WACV [Datasets Track](#) submission

Paper ID *****

001 1. Protocol Freeze and Governance

002 The seed campaign was governed by a frozen
003 specification (`configs/seed_campaign/`
004 `SEED_CAMPAIGN.json`) and a decision log
005 (`SPATIAL_REASONING_DECISION_LOG.md`). Key
006 entries: the battery drift correction (2026-08-11) reinstate
007 the committed legacy Tier-A/B/C drivers; the training
008 deadlock incident (2026-08-12) and its non-recurrence on
009 fresh boxes; the campaign completion entry (2026-08-14)
010 with framing decisions; and the numerical-audit/claim-
011 hierarchy/literature-audit entries (2026-08-14).

012 2. Frozen Metric Definitions

013 From the committed analyzer (`analyze_tier.c.py`),
014 read as the authoritative definitions:

- 015 • $A_{\text{transform}}(m) = P(\text{transformed prediction} =$
016 $\text{expected transformed label})$: transformed-answer
017 accuracy.
- 018 • $C_{\text{pair}}(m) = P(\text{pair consistency})$: the model's two an-
019 swers on the same example obey the linked-answer law.
020 `hflip_flip`: $P(\text{mirrored} \neq \text{normal})$ (answer-update rate);
021 `hflip_invariant`: $P(\text{mirrored} = \text{normal})$ (response-stability
022 rate); `relcomp/facingcomp`: flip-law compliance. Invalid
023 outputs count as non-consistent.
- 024 • $\text{both_correct}(m) = P(\text{normal-correct} \wedge$
025 $\text{transformed obeys the law})$: joint correctness.
- 026 • $G = A_{\text{normal}} - A_{\text{shuffle}}$ (correct-image dependence); $\Delta G =$
027 $G_{\text{tuned}} - G_{\text{zero-shot}}$.

028 3. Independent Numerical Audit (verbatim 029 summary)

030 `scripts/audit_paper2_numbers_independent.py`
031 recomputes every headline quantity from raw prediction
032 CSVs/JSONL and frozen manifests without importing any
033 analyzer module. Hard-fail rules: deterministic proportion
034 discrepancy $> 10^{-6}$; row-count mismatch; ID mismatch;
035 terminology mismatch. Verdict: PASS (0 issues, 0 failed
036 claims). Committed seed-0 reproduction: 7B normal
037 0.82414579 (diff 0.0); 2B normal 0.76492027 (diff 0.0).

Claim-level audits: ΔA positive all fresh seeds both
backbones; ΔG positive all fresh seeds both backbones;
fresh-seed C_{pair} within 0.05 of legacy General; all fresh 2B
`hflip` $C_{\text{pair}} > 2B$ zero-shot; Qwen3-VL-8B scoped. Hostile
tests (single-prediction flips) produced FAIL as expected
and PASS after restore.

044 4. Canonical Subset Sizes

normal/shuffle/blank/text_only $n=2195$; `relcomp` $n=666$;
facingcomp $n=103$; `hflip_flip` $n=245$; `hflip_invariant`
 $n=421$. All verified against frozen eligible-IDs manifests.

048 5. Archived Artifacts

All raw predictions, analysis JSONs, check-
points, training logs, figures, and audit outputs
are archived under `results/seed_campaign/`
in the repository (freeze commit 95b66bc on
`research/spatial-grounding-audit`).